Suhail Shaik Mohammed

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Software Engineer specializing in **Generative AI**, **NLP**, and **Full Stack Development**. Experienced in building RAG pipelines, fine-tuning BERT models, and deploying scalable web applications. Strong background in UI/UX Design andvAI (AI), aiming to deliver intelligent digital solutions.

Technical Skills

AI GenAI: RAG (Retrieval-Augmented Generation), BERT/Transformers, NLP, SHAP (Explainable AI), TensorFlow
Web Development: React.js, Tanstack, Node.js, Spring Boot, Flask, RESTful APIs, Tailwind CSS
Cloud DevOps: AWS (EC2, S3), Docker, Kubernetes, Git/GitHub
Design Tools: Figma, Adobe Illustrator, Canva, Wireframing, UI/UX Prototyping
Languages: Python, Java, JavaScript, C, Typescript, SQL

Projects

MINE – AI-Powered Social Platform | *React, RAG, AWS, Node.js* **mineglobal.org**
- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline to enable intelligent semantic search and AI context retrieval.
- Built a scalable Full Stack architecture using **React** and **Node.js**, deployed on **AWS EC2** for high availability.
- Designed a comprehensive Admin Dashboard with granular controls for content moderation and user management.

Airlines Sentiment Analysis (NLP) | *Python, BERT, Streamlit* **GitHub**
- Developed an advanced NLP model using **BERT Transformers** to classify sentiment from unstructured Twitter data.
- Achieved **83% Accuracy** by fine-tuning the model on 10,000+ records and handling noise (slang/emojis).
- Deployed the solution via a **Streamlit** interface, visualizing confidence scores and sentiment trends in real-time.

Predictive Crop Recommendation Engine | *Python, Random Forest, SHAP*
- Implemented a Random Forest model achieving **98% Accuracy** to predict optimal crops based on soil/climatic parameters.
- Integrated **SHAP (Explainable AI)** to provide transparent reasoning behind predictions, crucial for user trust.
- Designed a clean web interface for farmers to input real-time data and receive instant, data-driven suggestions.

Context-Aware Movie Recommendation System | *React, Flask, NLP* **GitHub**
- Developed a natural language search engine (e.g., "Show me inspiring dramas") using TF-IDF and Cosine Similarity.
- Architected a decoupled system with a React.js frontend and, integrating the TMDB API.
- Implemented mood detection logic to bridge the gap between user intent and static metadata genres.

Experience

Virtusa **Aug 2025 – Jun 2026**
Intern - Full Stack Engineering *Remote*
- Developing a robust **Inventory Management System** using **Spring Boot** and **React.js**.
- Optimizing SQL queries and API response times, reducing data retrieval latency by **35%**.
- Collaborating with senior engineers to implement Role-Based Access Control (RBAC) and secure data handling.

SkillsHap **Dec 2023 – Jan 2024**
Design & Frontend Team Lead *Remote*
- Led the **UI/UX Design** and frontend development of an Ed-Tech platform, translating **Figma** prototypes into responsive React code.
- Defined the visual identity and design system, resulting in a consistent user experience and a **40% increase** in engagement.
- Managed a team of developers, conducting code reviews and ensuring pixel-perfect implementation of design mockups.

E-Cell, VIT **Mar 2022 – Sep 2023**
Core Committee Member & Design Lead *Vellore, India*
- **Design Leadership:** Spearheaded the visual identity and branding for university-level entrepreneurship summits; designed high-fidelity website mockups and marketing assets using **Figma** and **Adobe Illustrator**.
- **Event Management:** Orchestrated the end-to-end logistics for large-scale events attended by 500+ participants; managed vendor relationships and coordinated with a team of 20+ volunteers.
- **Digital Presence:** Maintained the official club website and designed engaging social media content, increasing digital reach by 25%.

Education

Vellore Institute of Technology **Vellore, Tamil Nadu**
M.Tech. (Integrated) in Computer Science and Engineering; *Sept 2021 – June 2026*
- **Relevant Coursework:** Web Development, Design, Full Stack, Artificial Intelligence, Deep Learning, NLP, Predictive Analytics, Database Management.

Certifications

- **Microsoft Certified: Azure AI Fundamentals** (2024)
- **Data Science Certified (CodeWithHarry)** (2026)

Publications

- **AI-Driven Vision-Based Fitness Coaching Using Adaptive Skeleton Matching**
  IEEE Xplore